

Ismail Aissaoui
Chlef, Algeria
Phone: +213 660 70 77 96
Email: ismail.aissaoui.pro@gmail.com

June 16, 2026

Dr. Alexander Ziller
Institute for AI and Informatics in Medicine
Technical University of Munich (TUM)
Munich, Germany

Dear Dr. Ziller,

I am writing to express my strong interest in the PhD Candidate position in Trustworthy AI for Medicine within the Institute for AI and Informatics in Medicine under the direction of Prof. Daniel Rueckert. As an AI & Data Science Engineer with an extensive background in PyTorch and medical deep learning architectures, I am deeply motivated to contribute to your three-year research project focusing on privacy-preserving methods and synthetic medical data generation.

During my engineering and master's theses, I developed a strong foundation in addressing complex vulnerabilities and computational bottlenecks in medical AI. My recent work, "Cardiovascular Medical Digital Twin," successfully implemented a physics-informed digital twin for intensive-care patients to forecast Mean Blood Pressure (MBP) using 100Hz VitalDB ICU telemetry. I engineered a linear-time **Medical_MambaTS** framework for sub-millisecond bedside inference and a 0D Windkessel **PI_DeepONet** that utilizes PyTorch Autograd to enforce cardiovascular fluid dynamics.

Crucially, my experience addressing real-world medical data vulnerabilities—such as designing a custom "Missingness-Aware Gate" to handle sensor dropouts without corrupting the model state—aligns perfectly with your goal of investigating realistic vulnerabilities in deployed AI systems. I am highly motivated by your focus on bridging the gap between empirical privacy methods and provably robust solutions, as I have personally experienced the critical need for robust, generalizable AI in the healthcare sector where data privacy and fidelity are paramount.

I am eager to apply my strong mathematical foundations and PyTorch expertise to benchmark existing privacy-preserving methods and develop novel approaches with formal privacy guarantees. Working in a focused research team at TUM University Hospital is an environment where I know my independent research capabilities will thrive.

Enclosed are my CV, university transcripts, and a brief description of my Cardiovascular Medical Digital Twin project. Thank you for your time and consideration. I would welcome the opportunity to discuss my qualifications further in a brief introductory conversation.

Sincerely,

Ismail Aissaoui